

Research Proposal

Trustworthy and Explainable Large Language Models for Healthcare Decision Support

Abisha Jerlin Edison Prakash | University of Surrey | TrustMedLLM Research Project

Abstract

Artificial intelligence (AI) has transformative potential in clinical medicine, yet widespread adoption of AI-based clinical decision support systems (CDSS) remains constrained by three critical failures: hallucination of factually incorrect medical information [19], opaque black-box reasoning that clinicians cannot interrogate and static knowledge representations that cannot keep pace with the rapid growth of biomedical literature. TrustMedLLM is an end-to-end clinical decision support system designed to overcome the critical AI barriers of medical hallucinations, “black-box” reasoning and static knowledge. By integrating the BioBERT [1] encoder with the Retrieval-Augmented Generation (RAG) [2] module embedded in the PubMed [18] corpus, the framework ensures that outputs are based on real-time, evidence-based literature. To ensure transparency, the system employs SHAP [3,21] for global and instance-level interpretation, while conformal prediction [5,22] provides statistically calibrated uncertainty estimates. Together, these mechanisms reduce overconfidence and provide physicians a reliable, interpretable and factually consistent tool for making health care decisions.

1. Vision

1.1 Technical Challenges Addressed

The deployment of large language models in healthcare confronts four deeply interconnected technical challenges that jointly prevent safe clinical adoption. These challenges define the core scientific contribution of this proposal.

Knowledge Grounding and Hallucination Mitigation: Standard LLMs often generate plausible-sounding but factually incorrect medical information. We address this by using Retrieval-Augmented Generation (RAG) to anchor model outputs to peer-reviewed evidence from PubMed.

Domain-Specific Linguistic Nuance: Generic models struggle with complex clinical nomenclature and abbreviations. By utilising BioBERT, we address the challenge of specialised medical natural language understanding (NLU).

Black-Box Interpretability: Deep learning models cannot explain *why* a certain diagnosis was reached [21,25]. We integrate SHAP (global feature importance) to provide clinicians with a visual and mathematical breakdown of the most influential patient data points. SHAP gives more robust and global insights.

Rigorous Uncertainty Quantification: Clinical decisions require knowing when the AI is unsure. We implement conformal prediction [22] to move beyond simple "SoftMax" probabilities, providing a statistically guaranteed "confidence set" of potential diagnoses rather than a single, potentially overconfident guess.

1.2 Why this challenge needs to be addressed

Clinical decision support systems thus represent a frontier where AI can be responsibly harnessed to promote patient safety. Diagnostic errors contribute to over 250,000 preventable adverse events annually in the United Kingdom alone [6], representing one of the leading causes of patient harm. Despite rapid advances in machine learning, AI-driven CDSS adoption has remained frustratingly slow because clinicians cannot trust systems they cannot interrogate [7].

Indeed, the NHS Long Term Plan specifically sets out an ambition for AI to support clinical workflows and thereby speed up diagnosis and improve population health outcomes. [8] Meeting this mandate requires not merely accurate models but systems that are trustworthy, auditable, transparent and regulatory-compliant. There are more than 1 million new publications added to the global biomedical literature each year and therefore, no single clinician can realistically keep pace with the emerging evidence and its continuous integration into clinical practice [9]. An AI system capable of continuously grounding its reasoning in the latest peer-reviewed evidence represents a fundamental paradigm shift in evidence-based medicine [10].

Indeed, the current opaqueness of AI models perpetuates and further compounds existing health inequalities. Biased training data produces biased recommendations, disproportionately harming marginalised patient groups and without interpretability, these biases cannot be detected or corrected [11].

1.3 Limitations of Existing Work

Current clinical AI systems are hampered by three "black box" problems: The problem of hallucinations [19]: Generic LLMs are not based on real-time medical facts, often fabricating supplements or treatments that are dangerously out of date. Interpretability gap: Most models provide a "yes/no" diagnosis without specifying the cause. Even when tools such as SHAP [3] are used, they are often applied in isolation and lack clinical context. The darkness of uncertainty: Models are "overconfident". They will give a diagnosis even if the patient data is completely new to them, failing to indicate when a human doctor should intervene [12].

1.4 Importance and Quality of Proposed Work

TrustMedLLM represents the system design that integrates four models; BioBERT [1] gives you the domain understanding. RAG [2] is based on evidence grounding, SHAP [3] for explainability and conformal prediction [5] on uncertainty quantification, with a clinically validated framework.

The primary contribution of this work is not only integrating a single module; it integrates into a unified architecture into a coherent deployment system. The quality of the proposed work is ensured through comprehensive evaluation on biomedical benchmarks, including the PubMed clinical database. This brings together expertise in biomedical NLP, clinical informatics, trustworthy AI and healthcare ethics, ensuring real-world clinical validity [13].

1.5 Feasibility and Suitability

The feasibility of the proposed work is open-access technologies. BioBERT is available via HuggingFace Transformers [1]; PubMed provides a well-documented API supporting millions of queries per day. SHAP has robust, production-ready Python libraries [3,21] and conformal prediction is a framework that is available through the MAPIE library. Computational resources will utilise cloud GPU infrastructure, ensuring scalability.

1.6 Expected Outcomes

This project's key deliverables span a comprehensive set of technical and scholarly outcomes aimed at advancing trustworthy biomedical AI. On the technical side, this project will deliver an open-source TrustMedLLM pipeline that integrates BioBERT-based named entity recognition, retrieval-augmented generation (RAG), model explainability technique SHAP [21] and conformal prediction for uncertainty quantification.

1.7 Direct and Indirect Benefits & Beneficiaries

Direct Benefits

The system improves the accuracy and reliability of clinical decision support by reducing hallucinations and grounding outputs in verified medical evidence [19]. Explainable outputs enhance transparency, allowing clinicians to better understand and trust system recommendations.

Indirect Benefits

AI Healthcare Industry: Open-source TrustMedLLM provides a production-ready reference architecture for commercial CDSS developers [14].

International Health Systems: The framework is designed for generalisability across disease domains and healthcare systems, enabling global translation [15].

1.8 Impacts on Society, Economy and Environment

The implementation of advanced AI in healthcare has big benefits for three main areas: society, the economy and the environment. Explainable and auditable AI can help reduce health inequities by detecting racial and social economic biases in recommendations. More trust in healthcare providers accelerates out the safe use of these technologies in the NHS [16]. Finally, RAG-based inference essentially takes care of the environmental impact by getting rid of the need for expensive full-model retraining cycles. This cuts down on GPU energy use and the overall carbon footprint compared to traditional fine-tuning methods [17].

2. Approach

2.1 Objectives

The main objective of this project is to develop TrustMedLLM, a trustworthy and explainable large language model (LLM) framework for healthcare decision support. The system aims to reduce hallucinations, provide verifiable explanations, quantify uncertainty and ensure safety, fairness and clinical reliability through human oversight [7,31].

Objective 1: Develop a Retrieval-Augmented Generation (RAG) pipeline grounded in PubMed to ensure outputs are supported by verified biomedical evidence. This reduces hallucinations and improves factual accuracy [2,19,33].

Objective 2: Implement an evidence-based reasoning layer that produces citation-aware, clinically meaningful explanations alongside model predictions, enabling traceable and transparent outputs [2,20,27].

Objective 3: Enhance domain-specific understanding via BioBERT fine-tuning on retrieved biomedical documents, improving comprehension of complex medical terminology and context [1,34].

Objective 4: Integrate uncertainty quantification and explainability, including confidence scoring, SHAP-based interpretability and low-confidence flags, helping clinicians identify uncertain predictions [3,21].

Objective 5: Ensure trust, fairness and safety through conformal prediction, bias evaluation, safety validation and human-in-the-loop (HITL) monitoring, mitigating algorithmic bias, overconfidence and unsafe recommendations [5,7,13,23,24,28].

Together, these objectives address the main challenges of LLMs in healthcare-hallucinations, limited verifiability, domain knowledge gaps, overconfidence and bias-enabling a reliable and clinically useful decision support system.

2.2 Work Packages

The project is organised into five work packages, each addressing a critical component of the TrustMedLLM framework and defining clear technical responsibilities and deliverables.

WP1 – Retrieval and Knowledge Grounding

Objective: Develop a RAG retrieval pipeline over PubMed to ensure evidence-backed LLM outputs [2,33].

Methodology:

- Implement PubMed API access and document extraction.
- Chunk retrieved documents and create embeddings.
- Store embeddings in a vector database (FAISS/ChromaDB).
- Retrieve top-k relevant documents for each query.

Deliverables:

- Functional RAG retrieval pipeline.
- Indexed PubMed document library.
- Evaluation report of retrieval precision and recall.

Milestones:

- Successful retrieval of clinically relevant PubMed documents for test queries.

WP2 – Evidence-Based Reasoning Layer

Objective: Convert retrieved documents into structured, citation-aware responses to reduce hallucination and improve transparency [2,20].

Methodology:

- Integrate retrieved evidence with LLM outputs.
- Rank and filter evidence for clinical relevance.
- Generate citation-aware explanations referencing PubMed sources.

Deliverables:

- Reasoning module producing evidence-backed outputs.
- Evaluation of factual correctness and grounding.

Milestones:

- Demonstrated system outputs with accurate and traceable citations.

WP3 – Biomedical Language Understanding

Objective: Improve domain-specific understanding of biomedical text using BioBERT [1,34].

Methodology:

- Fine-tune BioBERT on retrieved PubMed documents.

- Enhance entity recognition and comprehension of clinical terminology.
- Integrate BioBERT into the RAG pipeline.

Deliverables:

- Fine-tuned BioBERT model for clinical tasks.
- Performance evaluation on NER and QA benchmarks.

Milestones:

- Improved biomedical text comprehension and entity recognition accuracy.

WP4 – Uncertainty and Explainability

Objective: Implement interpretable outputs with quantified uncertainty using SHAP [3,21].

Methodology:

- Add confidence scoring and low-confidence flags for outputs.
- Use SHAP to provide feature-level interpretability.
- Generate visual and textual explanations to help clinicians understand model decisions.

Deliverables:

- Explainability module with SHAP outputs.
- Confidence scoring and uncertainty assessment system.

Milestones:

- Interpretable outputs for each prediction with clear confidence measures.

WP5 – Trust & Reliability

Objective: Ensure fairness, safety and human oversight using conformal prediction and HITL mechanisms [5,7,13,23,24,28].

Methodology:

- Apply conformal prediction for calibrated uncertainty estimation.
- Evaluate demographic bias and conduct fairness assessments.
- Create HITL interface for clinician review of high-risk predictions.

Deliverables:

- Bias and safety evaluation report.
- Conformal prediction module with calibrated confidence measures for predictions.
- HITL review workflow for clinicians' supervision of critical outputs.
- Reliable prediction system integrating CP, fairness checks and HITL oversight.

Milestones:

- Fully validated and monitored prediction system integrating conformal prediction, fairness evaluation and human-in-the-loop oversight for safe clinical decision support.

2.3 Risk Mitigation Table

Risk	Impact	Likelihood	Mitigation Strategy
PubMed API rate limits restrict retrieval	Medium	High	Use request batching, caching, offline backups, institutional API keys [18,2,33].
BioBERT fine-tuning too slow or expensive	High	Medium	Use cloud GPU/HPC, LoRA fine-tuning, or frozen model with domain-adapted prompts [1,34].
HITL clinician availability limited	High	Medium	Use medical students or validated case vignettes as surrogate evaluators [23,24,31].
Residual hallucinations	High	Medium	Implement atomic fact-checking, abstain when confidence is low and leverage evidence-based retrieval [2,19,33].
Demographic bias not fully mitigated	High	Low	Counterfactual data augmentation, bias-aware reranking and fairness checks [11,28,16].
Team member unavailability	High	Low	Modular WP structure, assistant lead and shared repository [7,31].
Scope creep	Medium	Medium	Weekly check-ins, minimal feasible deliverables and strict milestones [31,32].

2.4 Ethical and Security Considerations

This project prioritizes ethical and secure use of AI in healthcare by addressing fairness, privacy, clinical safety, responsible usage and proper data handling. Demographic biases will be continuously evaluated and mitigation strategies such as bias-aware reranking and counterfactual data augmentation will ensure equitable performance across patient groups [7,11,16,28]. Privacy will be maintained by anonymizing datasets, collecting user data only with informed consent and following GDPR and UK Data Protection Act standards, with secure data handling including encryption, role-based access, audit logging and a structured data lifecycle covering collection, storage, processing and disposal [9,20]. The system is explicitly designed as a decision support tool, not a replacement for clinicians, with low-confidence outputs flagged for human review using a human-in-the-loop approach to prevent over-reliance on AI and measures in place to avoid misuse outside intended clinical applications [6,12,22,23,24]. To reduce hallucinations and improve reliability, evidence-based retrieval from PubMed and conformal prediction will provide calibrated and trustworthy outputs [2,5,19,33]. Input validation, output filtering and system robustness measures will ensure resilience against malicious inputs, adversarial attacks and other forms of abuse [6,20,31]. Bias and fairness will be continuously monitored using disaggregated evaluation metrics, with mitigation applied as needed, ensuring the system remains transparent, safe and ethically responsible while maximizing its practical impact in real clinical settings [6,9,30,35].

2.7 Dissemination Strategies

The results of this project are intended to be shared in ways that are accessible and useful for students and healthcare professionals. Research on improving clinical language models could be disseminated through journals and conferences in medicine and computer science [2,9,25,26,34]. The system could also be demonstrated through presentations, workshops and webinars so that clinicians and researchers can examine how it works in practice [23,24,29,35]. Important findings, including uncertainty handling, fairness and human oversight, can be communicated to clinical teams through clear reports and step-by-step guides [6,7,12,13]. Code and documentation can also be made available for academic and community use to support reproducibility and adoption [26,32,35].

Gantt Chart

Timeline of the TrustMedLLM project (September 2025-September 2027), showing active work periods for each work package. Shaded cells indicate when tasks are being actively worked on.

Deliverables		Timeline									
Part	Description	06/2025	09/2025	12/2025	03/2026	06/2026	09/2026	12/2026	03/2027	06/2027	09/2027
WP1	Retrieval and Knowledge Grounding										
1.1	Functional RAG retrieval pipeline.										
1.2	Indexed PubMed document library.										
1.3	Evaluation report of retrieval precision and recall.										
WP2	Evidence-Based Reasoning Layer										
2.1	Reasoning module producing evidence-backed outputs.										
2.2	Evaluation of factual correctness and grounding.										
WP3	Biomedical Language Understanding										
3.1	Fine-tuned BioBERT model for clinical tasks.										
3.2	Performance evaluation on NER and QA benchmarks.										
WP4	Uncertainty and Explainability										
4.1	Explainability module with SHAP outputs.										
4.2	Confidence scoring and uncertainty assessment for predictions.										
WP5	Safety, Bias, and Human-in-the-Loop										
5.1	Bias and safety evaluation report.										
5.2	Conformal prediction module with calibrated confidence measures for predictions.										
5.3	HITL review workflow for clinicians supervision of critical outputs.										
5.4	Reliable prediction system integrating CP, fairness checks and HITL oversight.										

Milestone Summary (2025–2027)

- M1 (Sept 2025):** RAG pipeline developed; PubMed library indexed and retrieval operational (WP1).
- M2 (June 2026):** Retrieval performance validated; precision and recall evaluation completed (WP1).
- M3 (Sept 2026):** Evidence-based reasoning layer deployed with citation-aware outputs (WP2).
- M4 (March 2027):** BioBERT fine-tuned and validated on clinical NER and QA benchmarks (WP3).
- M5 (June 2027):** Explainability and uncertainty modules implemented using SHAP and confidence scoring (WP4).
- M6 (Sept 2027):** Full system integration and validation completed, including bias checks, conformal prediction and HITL workflow (WP5).

References

All references are formatted in the IEEE citation style.

- [1] J. Lee et al., “BioBERT: a pre-trained biomedical language representation model for biomedical text mining,” *Bioinformatics*, vol. 36, no. 4, Sep. 2019, doi: <https://doi.org/10.1093/bioinformatics/btz682>.
- [2] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *arXiv.org*, Apr. 12, 2021. <https://arxiv.org/abs/2005.11401>
- [3] S. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” *arXiv:1705.07874 [cs, stat]*, Nov. 2017, Available: <https://arxiv.org/abs/1705.07874>
- [4] M. T. Ribeiro, S. Singh, and C. Guestrin, ““Why Should I Trust You?”: Explaining the Predictions of Any Classifier,” *arXiv.org*, Feb. 16, 2016. <https://arxiv.org/abs/1602.04938>
- [5] A. N. Angelopoulos and S. Bates, “A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification,” *arXiv.org*, Dec. 2022, Available: <https://arxiv.org/abs/2107.07511>
- [6] E. H. Shortliffe and M. J. Sepúlveda, “Clinical Decision Support in the Era of Artificial Intelligence,” *JAMA*, vol. 320, no. 21, pp. 2199–2200, 2018, doi: <https://doi.org/10.1001/jama.2018.17163>.
- [7] A. Rajkomar, J. Dean, and I. Kohane, “Machine Learning in Medicine,” *New England Journal of Medicine*, vol. 380, no. 14, pp. 1347–1358, 2019, doi: <https://doi.org/10.1056/nejmra1814259>.
- [8] NHS, “The NHS long term plan,” 2019, Available: <https://www.longtermplan.nhs.uk/>
- [9] E. J. Topol, “High-performance medicine: the Convergence of Human and Artificial Intelligence,” *Nature Medicine*, vol. 25, no. 1, pp. 44–56, Jan. 2019, Available: <https://www.nature.com/articles/s41591-018-0300-7>
- [10] F. Jiang et al., “Artificial Intelligence in Healthcare: Past, Present and Future,” *Stroke and Vascular Neurology*, vol. 2, no. 4, pp. 230–243, Jun. 2017, doi: <https://doi.org/10.1136/svn-2017-000101>.
- [11] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, “Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations,” *Science*, vol. 366, no. 6464, pp. 447–453, Oct. 2019, doi: <https://doi.org/10.1126/science.aax2342>.
- [12] E. Strickland, “IBM Watson, heal thyself: How IBM overpromised and underdelivered on AI health care,” *IEEE Spectrum*, vol. 56, no. 4, pp. 24–31, Apr. 2019, doi: <https://doi.org/10.1109/mspec.2019.8678513>.
- [13] H. Suresh and J. Gutttag, “A Framework for Understanding Sources of Harm throughout the Machine Learning Life Cycle,” *Equity and Access in Algorithms, Mechanisms, and Optimization*, Oct. 2021, doi: <https://doi.org/10.1145/3465416.3483305>.
- [14] A. M. Chekroud et al., “The promise of machine learning in predicting treatment outcomes in psychiatry,” *World Psychiatry*, vol. 20, no. 2, pp. 154–170, May 2021, doi: <https://doi.org/10.1002/wps.20882>.
- [15] NIHR, “AI in health and care award,” 2021.
- [16] L. A. Celi et al., “Sources of bias in artificial intelligence that perpetuate healthcare disparities—A global review,” *PLOS Digital Health*, vol. 1, no. 3, p. e0000022, Mar. 2022, doi: <https://doi.org/10.1371/journal.pdig.0000022>.
- [17] OpenAI, “GPT-4 technical report,” 2023, doi: <https://doi.org/10.48550/arXiv.2303.08774>.
- [18] National Library of Medicine, “PubMed,” 2024.

- [19] Z. Ji et al., “Survey of Hallucination in Natural Language Generation,” *ACM Computing Surveys*, vol. 55, no. 12, March 2023, doi: <https://doi.org/10.1145/3571730>.
- [20] J. Amann et al., “Explainability for artificial intelligence in healthcare: A multidisciplinary perspective,” *BMC Medical Informatics and Decision Making*, vol. 20, 2020.
- [21] A. Salih et al., “A perspective on explainable artificial intelligence methods: SHAP and LIME,” *arXiv preprint*, 2023.
- [22] J. C. Cresswell et al., “Conformal prediction sets improve human decision making,” in *Proc. ICML*, 2024.
- [23] X. Wu et al., “A survey of human-in-the-loop machine learning,” *arXiv preprint*, 2021.
- [24] J. Wang et al., “Human-in-the-loop machine learning: A macro-micro perspective,” *arXiv preprint*, 2022.
- [25] A. Esteva et al., “A guide to deep learning in healthcare,” *Nature Medicine*, vol. 25, pp. 24–29, 2019.
- [26] K. Rasheed et al., “Explainable, trustworthy, and ethical machine learning for healthcare: A survey,” *Computers in Biology and Medicine*, vol. 149, 2022.
- [27] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” *arXiv preprint arXiv:1702.08608*, 2017.
- [28] N. Mehrabi et al., “A survey on bias and fairness in machine learning,” *ACM Computing Surveys*, vol. 54, no. 6, pp. 1–35, 2021.
- [29] K.-H. Jung, “Large language models in medicine: Clinical applications, technical challenges, and ethical considerations,” *Healthcare Informatics Research*, vol. 31, no. 2, pp. 114–124, 2025.
- [30] I. Y. Chen et al., “Ethical machine learning in healthcare,” *Annual Review of Biomedical Data Science*, vol. 4, pp. 123–144, 2021.
- [31] C. J. Kelly et al., “Key challenges for delivering clinical impact with artificial intelligence,” *BMC Medicine*, vol. 17, no. 1, 2019.
- [32] OAE Publishing, “Applications of artificial intelligence in healthcare,” *Intelligent Robotics*, 2024.
- [33] Liu, Siru et al. “Improving large language model applications in biomedicine with retrieval-augmented generation: a systematic review, meta-analysis, and clinical development guidelines.” *Journal of the American Medical Informatics Association : JAMIA* vol. 32,4 (2025): 605-615. doi:10.1093/jamia/ocaf008
- [34] K. Singhal et al., “Large language models encode clinical knowledge,” *Nature*, vol. 620, pp. 172–180, 2023.
- [35] M. Mesinovic, P. Watkinson and T. Zhu, “Explainability in the age of large language models for healthcare,” *Commun. Eng.*, vol. 4, Art. 128, 2025.

Project Provenance

This document is a GitHub research edition of the TrustMedLLM proposal. The original proposal was prepared within a collaborative university coursework setting. This public edition is curated by Abisha Jerlin Edison Prakash and is published alongside her independently developed WP5 Trust & Reliability work, which focuses on conformal prediction, fairness evaluation, explainability and human-in-the-loop clinical review.